ENHANCED MULTI-SCALE ATTENTION FRAMEWORK

for Retinal Vessel Segmentation

Dataset-wise extensions over the LMFR-Net base architecture

RESEARCH PROPOSAL PRESENTATION

DRIVE
CHASE_DB1
STARE

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Study overview: one task, three dataset-specific pipelines

Scope, experimental framing and evaluation objectives

DATASETS

03

DRIVE • CHASE_DB1 • STARE

MODEL IMPLEMENTATIONS

03

LMFRNet_V2 • U-Net • Attention U-Net

EVALUATION METRICS

09

Overlap • classification • topology • boundary

Research focus

- Segment retinal vessels from colour fundus images into binary vessel masks.
- Use dataset-specific preprocessing and architecture choices instead of one shared pipeline.
- Evaluate region overlap, sensitivity/specificity, discrimination, connectivity and boundary distance.
- Compare newer run outputs with the original Base-vs-Ours results in the supplied proposal PDF.

How to read this deck

- Original PDF results are marked as proposal-PDF results.
- New DRIVE and CHASE figures are the values supplied for this update.
- STARE results use the values reported in the PDF, as requested.
- Sample-image metric strips are identified as sample outputs, not dataset averages.

The numbers describe different reported runs; cross-dataset scores should not be treated as a controlled head-to-head benchmark without matched splits and protocols.

The three datasets — what they contribute

All three datasets share the same target: fundus image → binary vessel mask

DRIVE

Digital Retinal Images for Vessel Extraction

SIZE	**40 images**
SPLIT / SUBSET	20 train / 20 test
RESOLUTION	565 × 584
POPULATION	Adults
ANNOTATIONS	Standard benchmark

CHASE_DB1

Child Heart and Health Study in England

SIZE	**28 images**
SPLIT / SUBSET	14 children (both eyes)
RESOLUTION	999 × 960
POPULATION	Pediatric
ANNOTATIONS	2 manual annotations / image

STARE

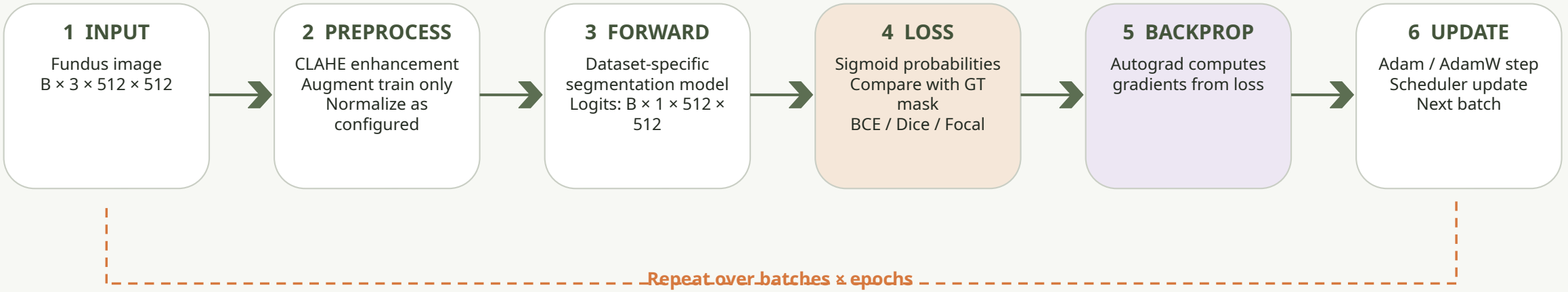
Structured Analysis of the Retina

SIZE	**20 images**
SPLIT / SUBSET	Vessel-segmentation subset
RESOLUTION	605 × 700
POPULATION	Mixed
ANNOTATIONS	2 expert annotations

Why three datasets? Different populations, image character and annotation protocols provide a broader test of generalization than a single benchmark alone.

End-to-end training loop

Batch-level flow used across all three datasets; implementation details vary by dataset



Input & preprocessing

- Input tensors are resized to 512 × 512.
- CLAHE improves local contrast; STARE applies CLAHE on the LAB L channel.
- Training augmentations are dataset-specific; validation/test images are not augmented.

Optimization & evaluation

- The model predicts one vessel-probability map per input image.
- Loss compares the probability map against the ground-truth vessel mask.
- Test predictions are thresholded at 0.5 in the supplied evaluation workflow.

Architecture map — why each dataset differs

The proposal uses separate architecture/training choices for DRIVE, CHASE_DB1 and STARE

DRIVE

LMFRNet_V2

ARCHITECTURE HIGHLIGHTS

- Residual multi-scale block: 1×1 / 3×3 / 5×5 / 7×7 kernels
- Attention gate weights encoder features using decoder signal
- Deep supervision via encoder–decoder skip

TRAINING CONFIGURATION

- AdamW (1e-4) • BCE + Dice
- H-flip + CLAHE • batch 4 • 100 epochs

CHASE_DB1

U-Net

ARCHITECTURE HIGHLIGHTS

- DoubleConv encoder / decoder with four skip connections
- Spatial resolution halves down the encoder and doubles up the decoder
- Bottleneck: 1024 × 32 × 32

TRAINING CONFIGURATION

- Adam (1e-3) • Cosine Annealing
- BCE + Dice • mixed precision ON
- H-flip + V-flip + rotation + CLAHE

STARE

Attention U-Net

ARCHITECTURE HIGHLIGHTS

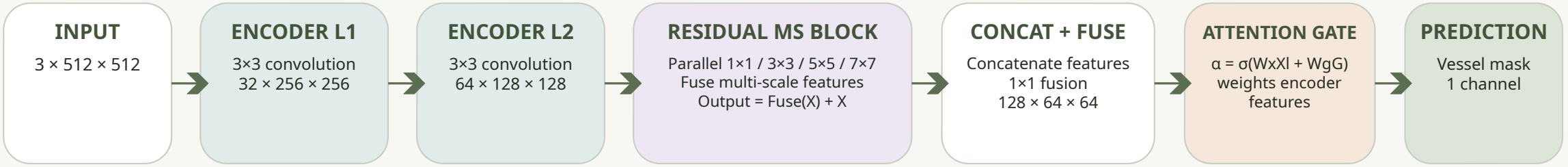
- LAB decomposition; CLAHE applied on L channel only
- Two attention gates modulate encoder skip features
- Deepest path passes through a 256 × 64 × 64 bottleneck

TRAINING CONFIGURATION

- AdamW (2e-4) • Cosine Annealing
- Focal + Dice ($\alpha = 0.8$, $\gamma = 2$)
- Weight decay 1e-4 • gradient clip 1.0

DRIVE — LMFRNet_V2 data flow

Multi-scale feature extraction fused with attention-gated skips



Key data-flow points

- Four kernel sizes run in parallel inside the residual multi-scale block.
- Residual shortcut adds the input feature to the fused output.
- Attention weights encoder features using a decoder gate signal.
- Encoder–decoder skip pathways preserve fine vessel detail.

Training configuration from the PDF

- Optimizer: AdamW (1e-4) • loss: BCE + Dice.
- Augmentation: horizontal flip + CLAHE.
- Batch size: 4 • epochs: 100.
- Input tensors are represented at 512 × 512 for the pipeline diagram.

Architecture and configuration are transcribed from the supplied proposal PDF (slides 4–5).

DRIVE — latest reported evaluation

These values are the updated DRIVE results supplied in the request, not the proposal-PDF Base-vs-Ours table

ACCURACY
0.9684
classification

DICE
0.7601
region overlap

IOU
0.6134
intersection / union

SENSITIVITY
0.7946
vessel recall

SPECIFICITY
0.9801
background recognition

Metric	Reported value	Interpretation
AUC	0.9818	Ranking / discrimination
MCC	0.7448	Balanced correlation measure
clDice	0.7762	Vessel connectivity / topology
HD95	7.1248	95th-percentile Hausdorff boundary distance
InferenceTime	0.0038	As reported by project team; unit not specified

Interpretation note: accuracy and specificity are high, while Dice / IoU and boundary distance show that vessel overlap and boundary precision remain the key improvement areas.

DRIVE — original proposal-PDF results & insights

Kept separate from the updated run reported on the previous slide

Metric	Base	Ours (PDF)
Accuracy	0.9658	0.9735
F1 / Dice	0.8237	0.8012
Recall / Sensitivity	0.8479	0.8120
Specificity	0.9830	0.9890

Extended metrics (Ours, PDF)

IoU 0.6675 • AUC 0.9725 • MCC 0.7810

clDice 0.8250 • HD95 4.8

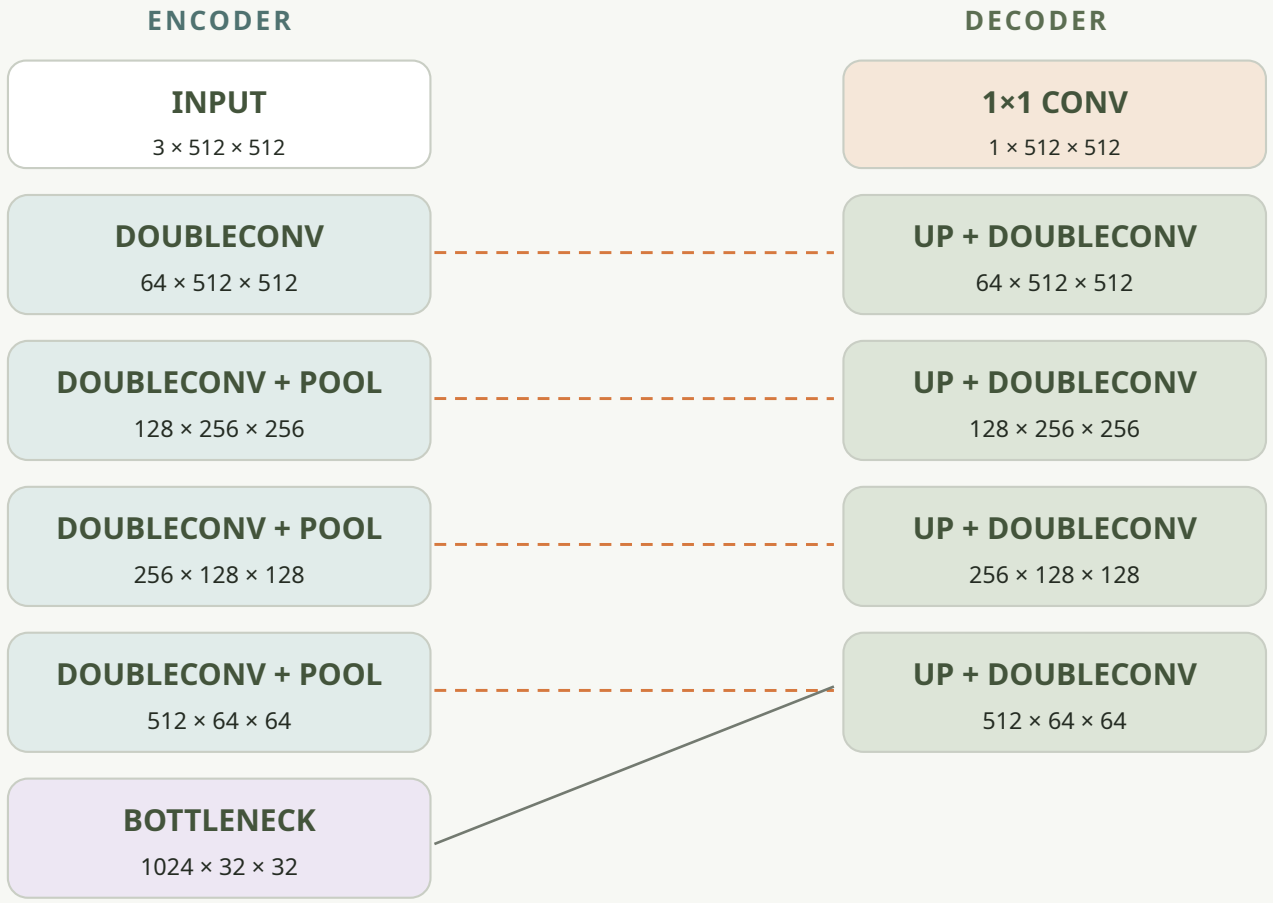
Practical takeaways reported in PDF

- Best accuracy gain: +0.77 percentage points (0.9658 → 0.9735).
- clDice 0.8250 was highlighted as the strongest connectivity result.
- HD95 = 4.8 was described as the sharpest boundary result.
- The proposal notes a Dice / IoU trade-off despite stronger accuracy and specificity.

Source: supplied PDF, "DRIVE — Results & Insights" (page 5).

CHASE_DB1 — U-Net layer-by-layer data flow

Four encoder–decoder skip connections preserve small-vessel detail



CHASE training setup

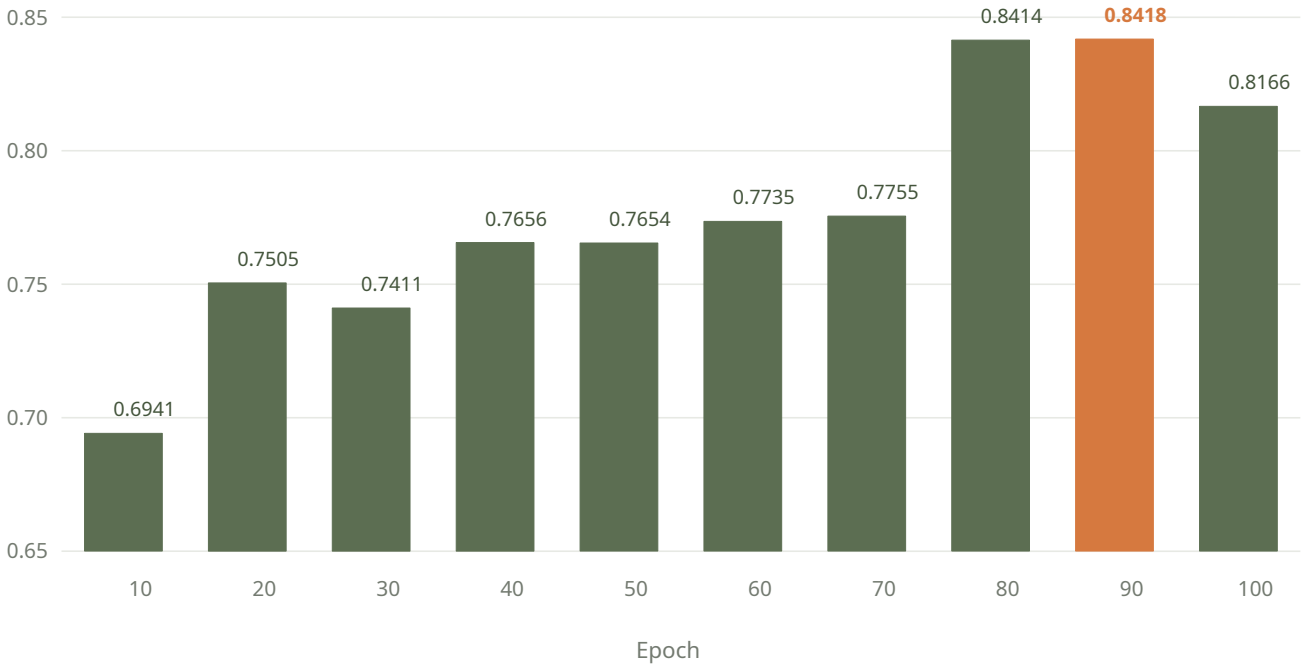
- Optimizer: Adam (1e-3).
- Scheduler: Cosine Annealing.
- Loss: BCE + Dice.
- Mixed precision: enabled.
- Augmentation: horizontal flip, vertical flip, rotation and CLAHE.
- DoubleConv block: Conv 3×3 → BN → ReLU → Conv 3×3 → BN → ReLU.

The PDF describes four encoder–decoder skip connections; dashed orange lines indicate skip pathways in this simplified diagram.

CHASE_DB1 — checkpoint history

Reported checkpoint values across 100 epochs; scores are reproduced exactly as provided

Saved-checkpoint Dice



HD95 by saved checkpoint

Epoch	HD95
10	13.45
20	9.00
30	4.47
40	9.85
50	14.21
60	12.04
70	11.05
80	2.00
90	5.00
100	8.49

Peak logged Dice: 0.8418 at epoch 90

Final evaluation is reported separately on the next slide (Dice 0.8351); best checkpoint Dice and final evaluation Dice are not the same statistic.

CHASE_DB1 — final evaluation results

Updated results and model-complexity values supplied by the project team

ACC	DICE	IOU	SEN	SPE
0.9726	0.8351	0.7169	0.8130	0.9875
final evaluation	final evaluation	final evaluation	final evaluation	final evaluation

AUC	MCC	CLDICE	HD95
0.9808	0.8206	0.8124	1.0000
final evaluation	final evaluation	final evaluation	final evaluation

PARAMETERS

8.36M

Model size

COMPUTE

207.90 GFLOPs

Reported model complexity

READ WITH THE CHECKPOINT LOG

Best logged Dice = 0.8418 (epoch 90); final evaluation Dice = 0.8351.

HD95 is kept at the value provided (1.0000); interpretation depends on the distance units, image scaling and evaluation implementation.

CHASE_DB1 — original proposal-PDF results & insights

Kept separate from the updated final evaluation on the previous slide

Metric	Base	Ours (PDF)
Accuracy	0.9740	0.9714
F1 / Dice	0.8291	0.7680
Recall / Sensitivity	0.8385	0.7523
Specificity	0.9844	0.9862

Extended metrics (Ours, PDF)

IoU 0.6240 • AUC 0.9270 • MCC 0.7536

clDice 0.7900 • HD95 5.5 (from the extended-results line)

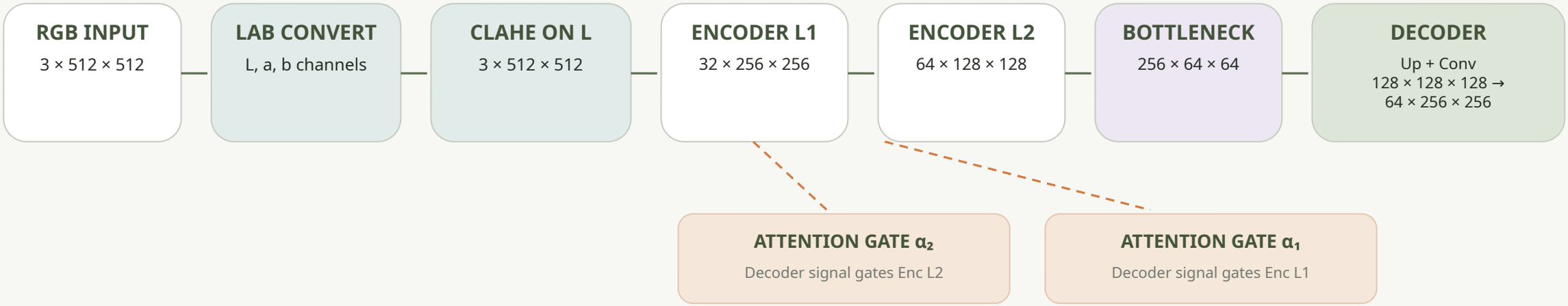
Insights recorded in the PDF

- Described as the hardest dataset for the pipeline.
- The PDF reports an F1 drop of 6.11% and a recall drop of 8.62%.
- Only 18 training images were available in the reported split.
- Boundary distance improved from HD95 6.0 to 5.5; specificity increased by 0.18 percentage points.
- The stated take-away is that F1 still needs tuning.

Source: supplied PDF, "CHASE_DB1 — Results & Insights" (page 7).

STARE — Attention U-Net data flow

LAB contrast enhancement with attention gates on encoder–decoder skip connections



Key data-flow points

- CLAHE operates on the L channel after RGB → LAB conversion.
- Attention weights α multiply encoder features before decoding.
- The deepest path passes through the bottleneck representation.

STARE training setup from the PDF

- AdamW (2e-4) + Cosine Annealing.
- Focal + Dice loss: α = 0.8, γ = 2.
- Weight decay 1e-4; gradient clipping 1.0; CLAHE only.

STARE — results & insights from the proposal PDF

STARE results below are the original proposal-PDF run, as requested

Metric	Base	Ours (PDF)
Accuracy	0.9699	0.9610
F1 / Dice	0.8257	0.7541
Recall / Sen	0.8685	0.8071
Specificity	0.9876	0.9734

Extended metrics (Ours, PDF)

IoU 0.6086 • AUC 0.9647 • MCC 0.7388

clDice 0.7800 • HD95 5.9

Interpretation recorded in the PDF

- The proposal reports the largest drop versus Base on STARE.
- It suggests Focal-Dice over-penalized easy background pixels on the small 20-image subset.
- The stated recommendation is to revert to BCE-Dice with stronger augmentation.
- The screenshot output with Dice = 0.780 is shown separately as a sample output; it is not substituted for the PDF aggregate.

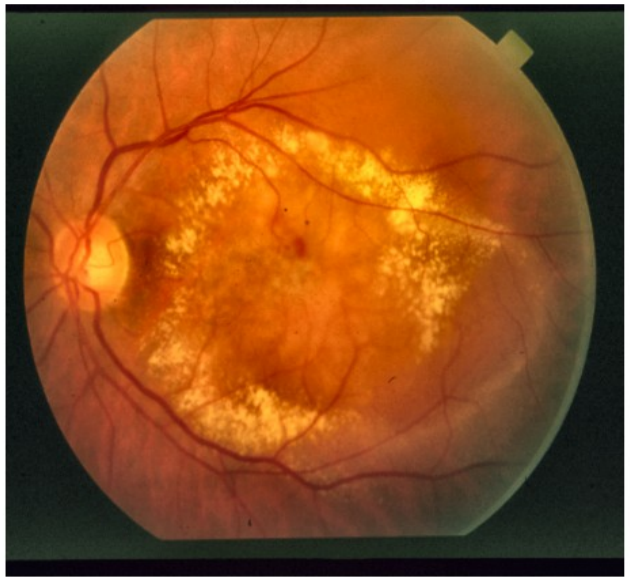
Source: supplied PDF, "STARE — Results & Insights" (page 9).

Qualitative output A — supplied segmentation snapshot

Original/enhanced fundus image, ground truth mask and prediction from the provided montage

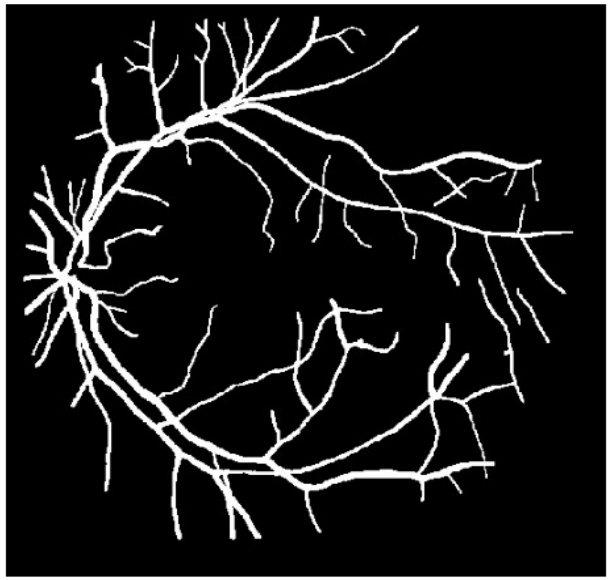
INPUT IMAGE

Original (Enhanced)



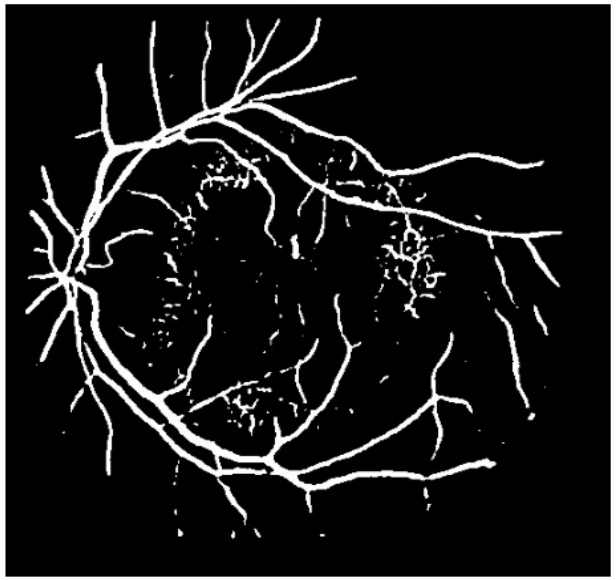
GROUND TRUTH

Ground Truth



PREDICTION

Prediction (Dice: 0.7277)



Metrics printed in the supplied montage: ACC 0.9540 | Dice 0.7277 | IoU 0.5720 | Sen 0.6900 | Spe 0.9798 | AUC 0.9196 | MCC 0.7039

These are the values shown on that sample output image; the updated DRIVE aggregate results are listed separately.

Qualitative output B — supplied segmentation snapshot

The source figure contains the input fundus, ground-truth vessels and prediction

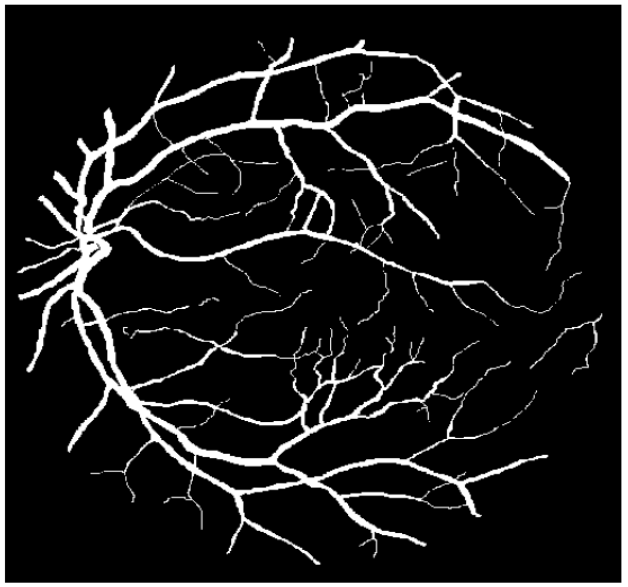
IMAGE

Image



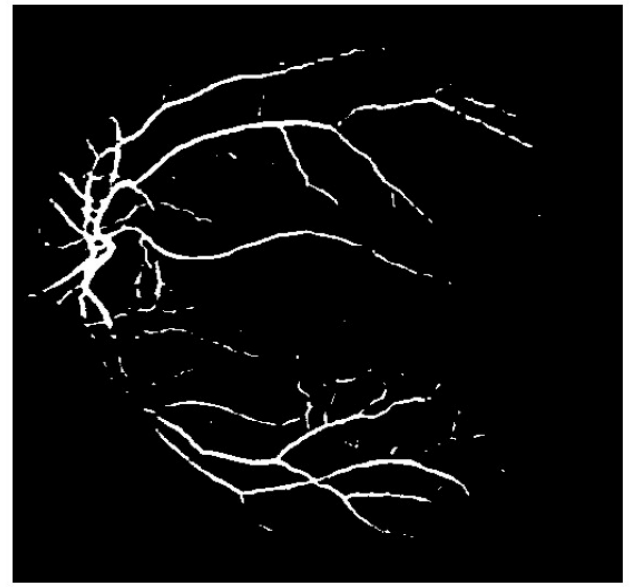
GT

GT



PREDICTION

Prediction



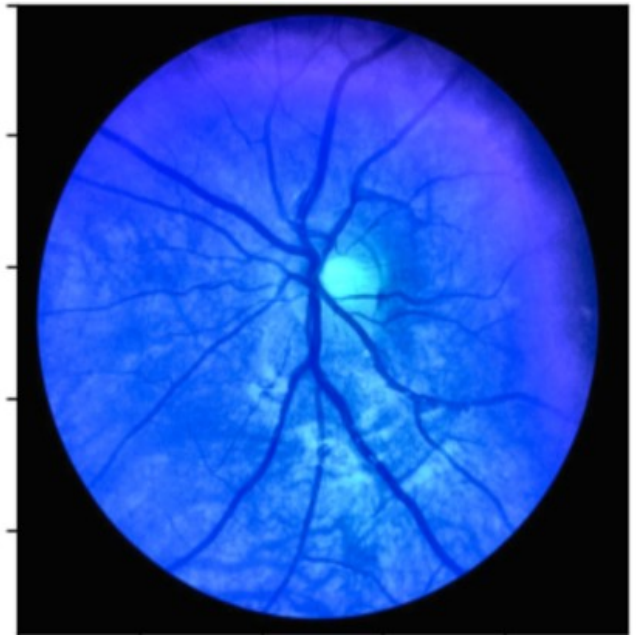
Qualitative comparison only — no dataset-level aggregate metrics are printed in this figure.

This image is presented as supplied and is not used to replace or recalculate any results table.

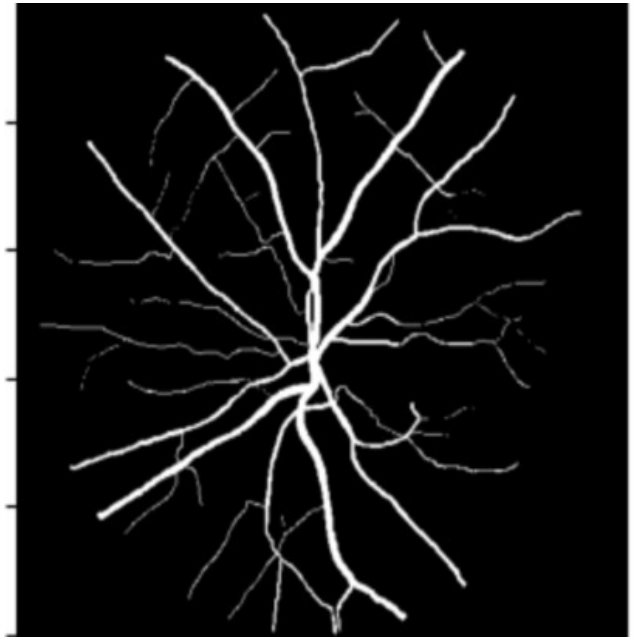
Qualitative output C — supplied screenshot

Retinal input, ground truth and the prediction displayed in the screenshot

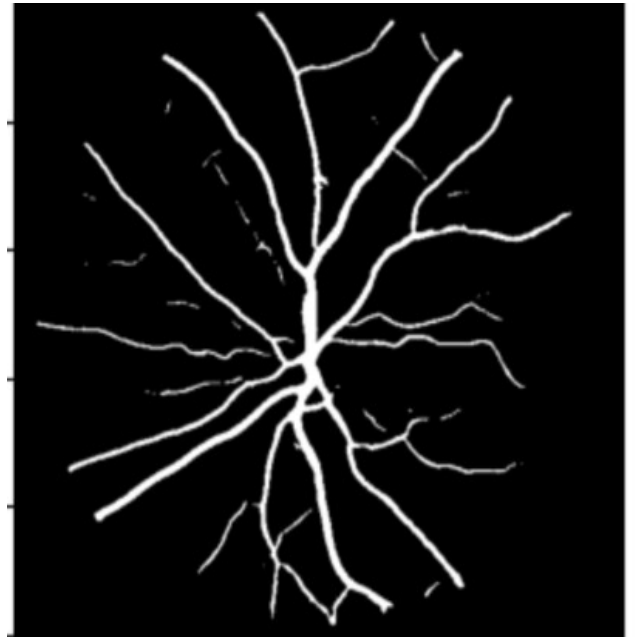
ORIGINAL



GROUND TRUTH



PREDICTION

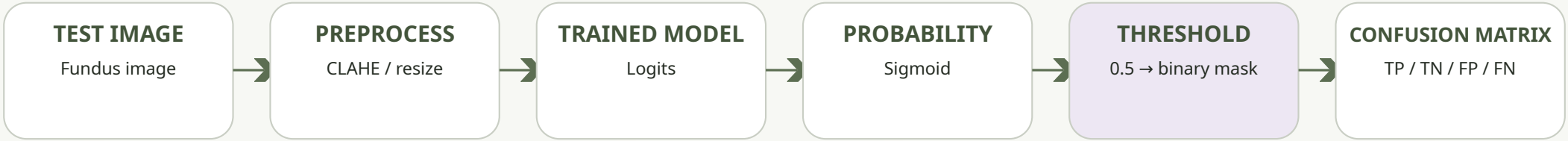


Dice shown in the supplied screenshot: 0.780

The displayed Dice belongs to the shown sample output; it is not assumed to be the dataset-wide mean.

Inference pipeline and metric interpretation

Nine metrics cover overlap, classification, discrimination, topology and boundary quality



Metric families

OVERLAP	**Dice, IoU**	How much predicted vessel area overlaps GT
CLASSIFICATION	**ACC, Sensitivity, Specificity**	Overall correctness, missed vessels, false positives
DISCRIMINATION	**AUC, MCC**	Probability ranking and balanced binary agreement
TOPOLOGY	**clDice**	Connectivity / broken branches
BOUNDARY	**HD95**	95th-percentile boundary distance

Evaluation checklist

- Thresholds and preprocessing must be held constant for fair comparisons.
- Report whether metrics are per-image averages or computed after pooling all pixels.
- HD95 depends on image scaling and physical units; report the implementation.
- A high accuracy can coexist with missed thin vessels because background pixels dominate.
- Sample images show local behavior; they do not replace aggregate test-set metrics.

Team work split and collaboration model

Dataset-wise ownership with shared research and evaluation decisions

MEMBER 1

DRIVE

MEMBER 2

CHASE_DB1

MEMBER 3

STARE

Collaborative contributions

- Architecture design — joint discussion and implementation review.
- Literature review — shared reading and synthesis.
- Evaluation protocol — unified definitions and reporting across datasets.
- Paper writing — section-wise division.
- Cross-dataset analysis — combined results after dataset-specific training.

Every member handled the full loop for one dataset:

Preprocessing → Training → Evaluation → Analysis

Team/member-to-dataset ownership is transcribed from the source PDF; member names were not mapped to individual datasets in that slide.

What was changed and implemented

Base-to-current changes reported by the supplied proposal PDF

Dimension	Base approach	Our implementation
Input	Raw image	CLAHE
Features	ICB only	+ Multi-scale features
Skip connections	LMAFF	+ Attention
Loss	CE + Dice	BCE-Dice and Focal-Dice (dataset-specific)
Optimizer	SGD	Adam / AdamW
Learning rate	Fixed	Cosine Annealing
Regularization	Weight decay	Weight decay + gradient clipping
Augmentation	Generic	Dataset-specific
Metrics	4 metrics	9 metrics

REPORTED BENEFITS

- +0.90% Sensitivity (DRIVE)
- +0.86% Sensitivity (STARE)
- HD95 = 4.8 (DRIVE)
- clDice 0.8250 (DRIVE)
- Attention aims to suppress false positives

Bottom line: the evaluation extends beyond F1 to topology and boundary precision, but overlap metrics still show dataset-specific trade-offs.

Cross-dataset result snapshot

Latest supplied DRIVE and CHASE runs; STARE uses the proposal-PDF Ours result as requested

Dataset / source	ACC	Dice	IoU	Sen / Recall	Spe	AUC
DRIVE — updated run	0.9684	0.7601	0.6134	0.7946	0.9801	0.9818
CHASE_DB1 — updated run	0.9726	0.8351	0.7169	0.8130	0.9875	0.9808
STARE — proposal PDF	0.9610	0.7541	0.6086	0.8071	0.9734	0.9647

Additional metrics retained per dataset

DRIVE (updated): MCC 0.7448 • clDice 0.7762 • HD95 7.1248 • inference time 0.0038.

CHASE (updated): MCC 0.8206 • clDice 0.8124 • HD95 1.0000.

Comparison caution

The three rows do not come from one controlled rerun: DRIVE/CHASE are the newly supplied metrics, while STARE remains from the source PDF. Use this table as a reporting snapshot, not a leaderboard.

STARE additional PDF metrics: MCC 0.7388 • clDice 0.7800 • HD95 5.9.

Conclusion — evaluate beyond a single score

The supplied results point to a balance between overlap, connectivity and boundary quality

DRIVE

LMFRNet_V2

- Updated run: Dice 0.7601; AUC 0.9818.
- PDF run highlighted accuracy +0.77 pp, clDice 0.8250 and HD95 4.8.

CHASE_DB1

U-Net

- Updated final Dice 0.8351; ACC 0.9726.
- Best logged checkpoint Dice 0.8418 at epoch 90; final test evaluation is separate.

STARE

Attention U-Net

- PDF Ours result: Dice 0.7541; HD95 5.9.
- PDF recommends revisiting Focal-Dice and strengthening augmentation for the small dataset.

CONSISTENT TAKEAWAY

Retinal vessel segmentation cannot be judged by one metric. Coverage (Dice / IoU), missed vessels (Sensitivity), false positives (Specificity), connectivity (clDice) and boundary distance (HD95) each reveal a different failure mode.

Recommended next steps: confirm identical split protocols, report whether metrics are macro-averaged or pooled, tune loss/augmentation on small datasets, and validate HD95 units and inference-time units.

THANK YOU

Questions & discussion

Enhanced Multi-Scale Attention Framework for Retinal Vessel Segmentation

Abhinav Pandey • Kritika Kumari • Aabhipsa Rajguru

Guided by Dr. Pujari Jeevana Jyothi | Department of CSE, VIT-AP University